

DXV — Data Exchange Vault

AML Data Ingestion · Feedback (Quarantine Export)

Pipeline	failed-record-export	Epic Key	DXV-FRE
Version	1.0	Author	Software Test Engineering
User Stories	10	Total Test Cases	62
Total Test Steps	175	Generated	2026-06-11

Suite Summary

Integration test suite for the DXV failed-record-export pipeline stage. This stage exports validation failures from Iceberg quarantine tables to MinIO as delimiter-separated files, together with acquisition success / error envelope flag files, using the same data contract and process_date as the wider AML flow. The suite below decomposes the stage into discrete functional and non-functional areas, each captured as an Agile User Story with executable, data-driven test cases.

In Scope

Covered by this suite

- Pipeline invocation, parameter resolution and configuration precedence.
- Reading Iceberg quarantine tables and per-table iteration / skip behaviour.
- Output path resolution, normalisation and ORG_UNIT prefixing.
- Generation of the three data exports (clean bad, aggregated, detailed).
- Envelope flag-file semantics (acq_success vs envelope_error).
- Export format, delimiter and header handling (csv/txt/pipe/gs).
- RFFilespec headerless alignment.
- Storage behaviour, .keep markers and *.zip.flag naming conventions.
- Downstream DXV integration and ingest-key alignment.
- Non-functional qualities: performance, reliability, security, compatibility.

Out of Scope

Explicitly excluded

- Upstream data-validation logic that places rows into quarantine.
- DXV-side archiving and routing of picked-up files.
- Outbound PGP encryption implementation (only its file-selection contract).
- Provisioning of MinIO / Iceberg infrastructure.

Test Environment

Object store	MinIO (flow-wide bucket + credentials)
Table format	Apache Iceberg quarantine tables
Runner	pipeline.run_pipeline("failed-record-export", ...)
Key env vars	ORG_UNIT (optional), WORKSPACE_ID, CATALOG
Process date	process_date in YYYYMMDD

User Story Index & Traceability

Story Key	Functional Area / Title	Epic Theme	TCs	
DXV-FRE-001	Invoke the failed-record-export pipeline with correct parameter resolution	Pipeline Orchestration & Configuration	7	
DXV-FRE-002	Read Iceberg quarantine tables and iterate contract tables safely	Quarantine Source Acquisition	6	
DXV-FRE-003	Resolve, normalise and prefix output object paths correctly	Output Layout & Object Keys	5	
DXV-FRE-004	Generate clean-bad, aggregated and detailed export files correctly	Data Export Content	6	
DXV-FRE-005	Emit the correct acquisition success / error envelope flag per run	Feedback Flags & Envelope Semantics	6	
DXV-FRE-006	Apply export file_format, separator and header rules correctly	Writer Format & Delimiters	8	
DXV-FRE-007	Align headerless export shape with RFFilespec ingest	RFFilespec / Headerless Compatibility	4	
DXV-FRE-008	Write .keep prefix markers and apply *.zip.flag naming conventions	Object-Store Behaviour & Naming	6	
DXV-FRE-009	Integrate exports with DXV pickup and align resubmit keys with ingest	Downstream Integration & Round-Trip	6	
DXV-FRE-010	Validate non-functional qualities of the quarantine export stage	Non-Functional Quality	8	

How to Use This Suite

Each User Story is delivered as a standalone Word (.docx) document and a parallel Markdown (.md) file with identical content. The Markdown files carry YAML front matter and are intended for automated import into Atlassian JIRA Cloud.

Execute the test cases in the order presented within each story. 'Happy Path' cases establish baseline behaviour; 'Negative', 'Boundary', 'Security' and 'Non-Functional' cases probe edges, failure handling and quality attributes.

Substitute the example test-data values (process_date, table names, paths, thresholds) with values appropriate to your environment and SLAs before execution.

Source: feedback-quarantine-export.md — generated 2026-06-11.